

UEIL Internship: Literature Review Of 12 Lead EKG Papers For Cardiac Arrhythmia Prediction

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Task: Need to do a literature review 12 lead EKG for cardiac arrhythmia prediction and select all features worth extracting for the 12 lead EKG.

MY FINDINGS - FEATURES ASSESSMENT

1. Temporal / Time-Domain Features

Standard time-domain features in ECG analysis include RR interval, P-wave duration, QRS duration, QT interval, heart rate variability (HRV), PR interval, and ST segment. Features which are worth extracting:

- *RR interval (time and distance measurements between successive R-peaks). Fluctuations or irregular variations are primary biomarkers for Atrial Fibrillation (AF) and ventricular ectopy.*
- *P-wave duration and amplitude*
- *PR interval (atrial conduction time) - Measures the time from the onset of atrial activation to the onset of ventricular activation. Essential for identifying Atrioventricular (AV) blocks.*
- *QRS duration (ventricular conduction time) - widened in bundle branch blocks. Prolonged QRS durations indicate conduction delays, such as Right or Left Bundle Branch Blocks (RBBB/LBBB)*
- *QRS count*
- *QT / QTc (QT Corrected) interval - the total time for ventricular depolarization and repolarization. Prolonged QTc is heavily linked to a high risk of Sudden Cardiac Death (SCD).*
- *ST slopes and segment elevation/depression*
- *T-wave amplitude and morphology*
- *R-peak amplitude*
- *Q onset and offset*
- *T offset*
- *R and T axis*

2. Heart Rate Variability (HRV) Features

HRV features include the following features:

- *atrial and ventricular rate (BPM)*
- *mean, median, standard/absolute deviation, variance and interquartile range of P waves, T waves, R waves and RR intervals*
- *standard deviation and variance of successive differences between RR intervals*
- *proportion of RR intervals*
- *geometric indices measuring triangular interpolation of the RR interval distribution with NeuroKit2 can generate 53 different HRV features per lead automatically*

3. Frequency-Domain Features

The frequency bands of 0.67–5 Hz, 1–7 Hz, and 10–50 Hz are commonly considered dominant components of P-wave, T-wave, and QRS-complex respectively. Key features here include:

- *Amplitude range (AR)*
- *Peak to peak amplitude (PPA)*
- *Mean amplitude (MA)*
- *Median Stepping Increment (MSI)*
- *Signal Integral (SI)*
- *Root mean Square (RMS) values of ventricular fibrillation*
- *Mean and Median Slope*
- *Smoothed Non-Linear Energy Operator (SNEO)*
- *Power Spectral Density (PSD) via Welch's method or FFT - P wave band (0.67-5 Hz) and QRS Power (10-50 Hz)*
- *Low-frequency (LF) and high-frequency (HF) power bands of HRV*
- *LF/HF ratio (sympathovagal balance)*
- *Dominant frequency per lead*

4. Features Analysis Tools

- *Spectral Energy Distribution*
- *Entropy (Shannon Weiner Entropy)*
- *Silhouette's score*
- *Davies-Bouldin Index*

- *Calinski–Harabasz Index*
- *Dunn Index*
- *Principal Component Analysis*
- *Isometric Analysis*

5. Time-Frequency Features (Wavelet-Based)

From the T-wave alone, 22 features can be extracted: 17 morphological and 5 wavelet features. Wavelet transforms give both temporal and frequency resolution simultaneously, which raw FFT cannot. Specific features which can be extracted are as follows:

- *Discrete Wavelet Transform (DWT) coefficients at multiple decomposition levels*
- *Continuous Wavelet Transform (CWT) for time-frequency maps (used in Paper 4's 2D CNN branch)*
- *Wavelet energy per sub-band*
- *P wave band (0.67-5 Hz)*
- *QRS Power (10-50 Hz)*

6. Morphological / Shape Features

Morphological features describe the geometric and amplitude characteristics of individual ECG waves, including QRS width, P–R and Q–T intervals, R-peak amplitude, and area under the QRS. Additional features worth extracting:

- *Area under QRS complex*
- *Amplitude of QRS complex*
- *Slope and standard deviation of ST segment - Measures the baseline shift after ventricular contraction. Key for identifying myocardial ischemia, infarction, or strain patterns.*
- *T-wave asymmetry and flatness - Represents ventricular repolarization anomalies, which assist in assessing hypertrophic cardiomyopathy and ventricular arrhythmia risks.*
- *P-wave presence/absence and amplitudes - Critical for predicting future incidences of Atrial Fibrillation. Significant clinical variations or declines in P-wave amplitude over time are independent indicators of AF risk.*

- *QRS axis (R-axis, T-axis from 12-lead spatial information) - Calculating the net vector of ventricular depolarization helps identify fascicular blocks and ventricular hypertrophy.*

7. Nonlinear / Statistical Features

Higher-order statistical features including skewness, kurtosis, and 5th-order moment of each signal are extracted as feature vectors, since their nonlinear nature helps highlight the dynamic aspects of ECG signals such as:

- *Sample entropy (signal complexity/regularity)*
- *Approximate entropy*
- *Fractal dimension (FD)*
- *Lyapunov exponent (LE)*
- *Approximate, sample, fuzzy and permutation entropy*
- *Poincaré plot descriptors (SD1, SD2) from HRV*
- *Extracting the skewness (asymmetry) and kurtosis (tailedness) of isolated segments provides advanced descriptions of subtle waveform deformities*

8. Multimodal / Clinical / Demographic Features

Clinically significant additions to the signal features such as the following:

- *Age and sex (strong independent predictors, especially for AF)*
- *BMI / obesity indicator*
- *Medication history (amiodarone, diuretics etc)*
- *Comorbidities (hypertension, diabetes)*

9. Deep Representation Features (End-to-End)

I found out that several papers skip hand-crafted features entirely and let CNNs/Transformers learn features directly from raw waveforms. The PhysioNet Challenge 2020 baseline used QRS amplitude and RR intervals from Pan-Tompkins detection combined with demographic features, but top-performing teams used end-to-end deep learning on raw signals.

10. Spectral Features

Examples of spectral features which could be used and extracted:

- *amplitude spectrum area - AMSA,*
- *centroid frequency - CF,*
- *peak frequency - PF,*
- *total energy - ENRG,*
- *spectral flatness measure - SFM,*
- *centroid power - CP,*
- *maximum power - MP,*
- *power spectrum analysis - PSA*

APPENDIX - LINKS AND ANALYSIS OF ALL PAPERS RESEARCHED

My Findings: Key foundational and recent papers in 12-lead ECG-based cardiac arrhythmia prediction emphasize **deep learning** and **discrete heartbeat analysis**.

These papers utilize **large datasets (such as PhysioNet and PTB)** to successfully predict conditions like **Atrial Fibrillation** and **Sudden Cardiac Death (SCD)**.

Key papers and their approaches include:

- **Discrete Heartbeat Analysis:** A landmark 2023 study by the NCBI utilized ResNet-18, LSTM, and Transformer models on 12-lead signals to identify subtle, early heartbeat patterns, successfully predicting Atrial Fibrillation and CIA-NSR without needing extra patient data. Access the study on NCBI PMC.
- **Atrial Fibrillation (AF) Risk Prediction:** Research published via the Nature portfolio focuses on forecasting AF up to 5 years in advance using standard 12-lead readings, improving early intervention and reducing misdiagnosis rates. Read the full details via NCBI PMC.
- **Sudden Cardiac Death Risk (SCD) in Genotype-Positive Patients:** A 2023 study highlights a deep learning-derived 12-lead ECG approach that outpaced traditional methods for assessing SCD risk in patients with hypertrophic cardiomyopathy. Explore the paper on NCBI PMC.

Meta-Analysis of 12-Lead ECG Classification: For a broad overview of the field, a meta-analysis evaluating over 40 teams' techniques from the CinC Challenge provides an excellent baseline on how computational algorithms classify and predict anomalies from 12-lead data. Review the comprehensive review on NCBI PMC.

Paper 1: Predicting Future Incidences of Cardiac Arrhythmias Using Discrete Heartbeats from Normal Sinus Rhythm ECG Signals via Deep Learning Methods

Link: [Predicting Future Incidences of Cardiac Arrhythmias Using Discrete Heartbeats from Normal Sinus Rhythm ECG Signals via Deep Learning Methods - PMC](#)

My Findings From The Paper - Key Points Worth Noting:

- Helped compare the effectiveness of using discrete heartbeat analysis vs 12 lead ECG as input to predict future cardiac arrhythmias and atrial fibrillation using deep learning models
- The utilization of discrete heartbeats as input data has been identified as a more optimal approach for detecting subtle abnormalities indicating future incidences of atrial fibrillation and other arrhythmias
- Experiments were conducted using two types of inputs: A combination of discrete heartbeats extracted from 12-lead ECG and an entire 12-lead ECG signal of 10s.
- This study utilized 326,904 ECG signals from 134,447 patients and categorized them into three groups: true-normal sinus rhythm (T-NSR), atrial fibrillation-normal sinus rhythm (AF-NSR), and clinically important arrhythmia-normal sinus rhythm (CIA-NSR).
- Clinically important arrhythmia included atrial fibrillation, atrial flutter, atrial premature contraction, atrial tachycardia, ventricular premature contraction, ventricular tachycardia, right and left bundle branch block, and atrioventricular block over the second degree. The AF-NSR group included normal sinus rhythm paired with atrial fibrillation or atrial flutter within 14 days, and the CIA-NSR group comprised normal sinus rhythm paired with CIA occurring within 14 days.
- Three deep learning models, ResNet-18, LSTM, and Transformer-based models, were utilized to distinguish T-NSR from AF-NSR and T-NSR from CIA-NSR.
- The analysis reveals that these discrete heartbeats contain subtle patterns that deep learning models can identify. Focusing on discrete

heartbeats may lead to more timely and accurate diagnoses of these conditions, improving patient outcomes and enabling automated diagnosis using ECG signals as a biomarker.

- Previous approaches to ECG analysis have mainly concentrated on using whole 12-lead EKG recordings as the input for deep learning models due to the popularity of two-dimensional CNNs in analyzing various data types, such as auditory signals that were transformed into two-dimensional image data.
- **Input Features: segmented single-beat waveforms across all 12 leads**

Paper 2: Atrial fibrillation risk prediction from the 12-lead electrocardiogram using digital biomarkers and deep representation learning

Link: [Atrial fibrillation risk prediction from the 12-lead electrocardiogram using digital biomarkers and deep representation learning - PMC](#)

My Findings From The Paper - Key Points Worth Noting:

- Used features obtained from different modalities, namely demographics, clinical information, engineered features, and features from deep representation learning.
- Also used 12 lead ECG biomarkers/morphology and heart rate variability (HRV) as features for the training set combined with deep learning features and representations (DNN as neural network feature representor for deep learning)
- 12 lead ECG recording can be used as a good feature due to its high representation of the training and validation data as biomarkers
- **Results of the paper indicate p values for features such as age, amiodarone, diuretics, sex, obesity, smoking etc without any personal information included**
- **Input Features: P-wave duration, PR interval, QRS duration, QT interval, and HRV metrics**

Paper 3: Diagnostic Accuracy of Artificial Intelligence for Arrhythmia Detection Using the 12-Lead Electrocardiogram: A Systematic Review and Meta-Analysis

Link: [Diagnostic Accuracy of Artificial Intelligence for Arrhythmia Detection Using the 12-Lead Electrocardiogram: A Systematic Review and Meta-Analysis | medRxiv](#)

My Findings From The Paper - Key Points Worth Noting:

- Used these features for dataset: first author's last name, year and country of publication, study design, study period, total number of patients, total number of ECGs, number of ECGs with and without arrhythmias in the test set, percentage of male patients, mean age, classes of arrhythmias in the study, AI architecture, standard of reference for diagnosis, accuracy, F1 score, and data necessary to calculate sensitivity, specificity, and AUC. If it was not possible to calculate the outcomes for all arrhythmias in the study, we extracted the data for **AF, which was the most common target label.**
- All studies analyzed standard 12-lead ECGs for arrhythmia detection or classification. Some focused on a single rhythm, most commonly AF, whereas others examined a broader arrhythmic spectrum, including atrial flutter, bundle branch blocks, premature complexes, supraventricular tachycardia, and ventricular tachycardia.
- **Standard Evaluation Metrics: Sensitivity, Specificity, AUC, F1 Score**

Paper 4: Automatic Classification Method of Arrhythmias Based on 12-Lead Electrocardiogram

Link: [Automatic Classification Method of Arrhythmias Based on 12-Lead Electrocardiogram](#)

My Findings From The Paper - Key Points Worth Noting:

- The model of this paper utilized electrocardiogram signals and a dual channel deep neural network to **extract different dimensional features from one-dimensional and two-dimensional electrocardiogram time–frequency maps, and combines attention mechanism to effectively fuse the important features of 12-lead**, thereby obtaining richer arrhythmia information and ultimately achieving accurate classification of nine types of arrhythmia signals.
- **Input Features: Extracted 13 features from the electrocardiogram signals using Gaussian Mixture Models and Wavelet features.**
- Found this important information in the paper:
 - *Current researches on automatic diagnostic models for arrhythmia have the following three main shortcomings.*
 - *Due to the limitations of data sets, most classification studies on arrhythmia are conducted on a single data set, and the proposed methods and models cannot be universalized or generalized.*
 - *Most of the researches working on automatic diagnosis of cardiac arrhythmias are based on modeling and analysis of single modal features extracted from one-dimensional ECG sequences, ignoring the frequency domain features of ECG signals.*
 - *Currently, many studies are based on single lead ECG signals, but 12-lead ECG can provide richer arrhythmia information. Different leads can reflect the electrical activity of the heart from different spatial angles, resulting in different leads presenting different shapes at the same time. How to effectively integrate the spatial characteristics of different leads is the focus of current model research.*
- This study has developed a multimodal feature fusion model based on attention mechanism, **which implements nine types of arrhythmia classification on a mixed dataset, including eight types of abnormal ECG and normal sinus rhythm. Multimodal**

features of ECG signals are extracted for classification of arrhythmia signals. A dual channel network is used to extract the one-dimensional time series features and two-dimensional time–frequency features of the 12-lead ECG signals, and then the feature fusion layer is used to integrate the two modal features.

- In this paper, a CNN is used as a feature extraction network. In order to efficiently learn the features in one-dimensional ECG sequences and two-dimensional ECG time–frequency maps, stacked one-dimensional and two-dimensional bottleneck residual structures are used as feature extraction networks. The bottleneck residual structure has the advantages of increasing network depth, enhancing feature transfer, and reducing the number of parameters. Therefore, using the bottleneck residual structure can improve the learning ability of deep neural networks.
- In this paper, the attention mechanism is combined with dual channel deep neural networks to preserve the useful information in one-dimensional ECG sequences and two-dimensional ECG time–frequency maps as fully as possible, reducing the loss of valuable information. For one-dimensional ECG sequences, introducing channel domain attention mechanisms can make one-dimensional ECG sequences pay more attention to the relationships between different channels. By aggregating the characteristic relationships between channels, the characteristic values between each channel can be adaptively adjusted. A mixed domain attention mechanism is introduced for two-dimensional time–frequency electrocardiograms, which enables the two-dimensional bottleneck residual structure to simultaneously enhance important features in both channels and spatial domains. That is, it can not only strengthen important channel features and suppress unwanted channel features, but also efficiently learn areas that require special attention in two-dimensional time–frequency electrocardiograms to preserve key features in the time–frequency diagram. Experiments have shown that introducing attention

mechanisms can better enable feature extraction networks to focus on key feature locations in the ECG band.

- **For the dataset collection, the paper focuses on classifying 9 classes of ECG signals, which are Atrial fibrillation (AF), First-degree atrioventricular block (IAVB), Left bundle branch block (LBBB), Right bundle branch block (RBBB), Premature atrial contraction (PAC), Premature ventricular contractions (PVC), Sinus Bradycardia (SB), Sinus Tachycardia (Stach), Normal (SNR).**
 - Key Idea:
 - Uses a dual channel network to extract one-dimensional time series features and two-dimensional time–frequency features of 12-lead ECG signals, and then use a feature fusion layer to integrate the two modal features.
 - Adds different attention mechanisms to the dual channel network to extract more discriminative features from different perspectives, that is, by giving different channels or regions different weights, flexibly capturing global and local relationships, allocating more attention to key feature areas, and allocating less attention to noise areas, thereby achieving enhancement of key features and noise reduction. The attention mechanism effectively reduces the interference of ECG signal noise on arrhythmia recognition, enabling the model to extract more robust features, and improving model classification performance.
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Paper 5: Cardiac Arrhythmia Detection With 12 Led ECG Using A Multimodal Neural Network

Link: [content](#)

My Findings From The Paper - Key Points Worth Noting:

- In the paper, the model was initially trained on data from the PhysioNet/Computing in Cardiology Challenges 2020 and 2021 using a signal-only convolutional neural network and analyzed with Shapley Additive Explanations to rank lead importance. It was then retrained and evaluated using 6, 4, and 2 leads, in both signal-only and multimodal configurations combining ECGs with demographic and temporal features.
- **Input Features: Incorporates patient data such as age, sex, symptoms etc. with all these parameters being fed into a neural network (features), with CNN and ResNet18.**
- Mostly ECG derived features from dataset which consisted of 12-lead ECG recordings sampled at 500 Hz with a duration of 10 seconds, accompanied by patient metadata and diagnoses with features from arrhythmic events and normal rhythm, where six arrhythmia classes were selected:
 - **atrial fibrillation (AF)**
 - **atrial flutter (AFL)**
 - **sinus arrhythmia (SA)**
 - **sinus bradycardia (SB)**
 - **sinus tachycardia (STach)**
 - **normal sinus rhythm (NSR)**
- **Age and gender were included as input features, with NeuroKit2 Python library used for ECG recordings and downsampled from 500 Hz to 250 Hz.** ECG leads and temporal features were extracted for this study.

Paper 6: Cardiac Arrhythmia Detection With 12 Led ECG Using A Multimodal Neural Network

Link: [Practical Lessons on 12-Lead ECG Classification: Meta-Analysis of Methods From PhysioNet/Computing in Cardiology Challenge 2020 - PMC](#)

My Findings From The Paper - Key Points Worth Noting:

- In the paper, I got these details:

1. **Temporal Features:** Temporal features are related to the morphological characteristics of ECG waves. The extraction of temporal features consists of two steps: wave detection and measurement computing. In Challenge 2020, teams usually employed traditional waves detection methods, such as **P-wave, QRS-complex, and T-wave, and then explicitly computed ECG measurements as feature vectors, such as P-wave duration, PR interval, QRS duration, and ST slope.** The details of the temporal features can be found in Hong et al. ([2019b](#)). These ECG-specific features have proven to be effective for the diagnosis of cardiac diseases.
2. **Frequent Features:** Frequency domain is also an important part of ECG hand features. Thus, some teams extracted features focusing on the frequency spectrum, excluding temporal information. **The frequency domain helps to inspect signals from a different view rather than only from the temporal domain. For example, the frequency bands of 0.67–5 Hz, 1–7 Hz, and 10–50 Hz are commonly considered as the dominant components of P-wave, T-wave, and QRS-complex, respectively.**

Paper 7: Automated detection of arrhythmias using a novel interpretable feature set extracted from 12-lead electrocardiogram

Link: [Automated detection of arrhythmias using a novel interpretable feature set extracted from 12-lead electrocardiogram - ScienceDirect](#)

My Findings From The Paper - Key Points Worth Noting:

- **Input Features:** The paper features are mathematical techniques such as the Fourier transform, wavelet transform, and cross-correlation for rigorous evaluation of ECG characteristics.

Paper 8: Arrhythmia Classification from 12-Lead ECG Signals Using Convolutional and Transformer-Based Deep Learning Models

Link: [2502.17887](#)

My Findings From The Paper - Key Points Worth Noting:

- Details from the abstract here:

*Within this purpose, we combined multiple datasets, usually used in the field of arrhythmias classification: PTB-XL electrocardiography dataset [41], PTB Diagnostic ECG Database [3], China 12-Lead ECG Challenge Database [24], Georgia 12-Lead ECG Challenge Database [13], and St. Petersburg INCART 12-lead Arrhythmia Database [37]. For the input data, we employed ECG signal processing methods, specifically a variant of the **Pan-Tompkins algorithm**, useful in arrhythmia classification because it provides a robust and efficient method for detecting QRS complexes in ECG signals. Additionally, we used machine learning techniques, widely used for the task of classification, including convolutional neural networks (1D CNNs, 2D CNNs, ResNet) and Vision Transformers (ViTs). The systems were evaluated in terms of accuracy and F1 score. We analyzed our dataset from two perspectives. First, we fed the systems with the ECG signals and the GRU-based 1D CNN model achieved the highest accuracy of 93.4% among all the tested architectures. Secondly, we transformed ECG signals into images and the CNN2D model achieved an accuracy of 92.16%.*

Paper 9: Arrhythmia Classification with Single-Channel Features Extracted from “A Large-Scale 12-Lead ECG Database for Arrhythmia Study”

Link: [Arrhythmia Classification with Single-Channel Features Extracted from “A Large-Scale 12-Lead ECG Database for Arrhythmia Study” - PMC](#)

My Findings From The Paper - Key Points Worth Noting:

- Details from the abstract here:

*This paper compares traditional morphological measures like **QRS duration, QT interval, atrial/ventricular rates with advanced time-, frequency-, and nonlinear-domain descriptors, using 15–39 features. In the four-class task, 15 MRMR-selected features achieved 94.2% accuracy.***

- From Introduction:

Numerical techniques include clustering and dispersion metrics, such as the Silhouette score [1], which evaluates the difference between intra- and inter-cluster distances for each sample; the Davies–Bouldin Index [2], which quantifies the ratio between intra-cluster dispersion and inter-cluster separation; and the Calinski–Harabasz Index [3], which reflects the ratio between global variance and within-cluster variance. Other relevant measures include the Dunn Index [3,4], which seeks to maximize the minimum inter-cluster distance while minimizing intra-cluster dispersion, and the Shannon–Wiener entropy index computed over local label neighborhoods, particularly informative in manifold-structured data.

Additionally, theoretical estimates of class overlap can be derived from distance-based measures such as Bhattacharyya, Mahalanobis, and Jeffries–Matusita distances, which characterize the divergence between probabilistic class distributions. From the perspective of feature interrelations, tools such as correlation matrices, mutual information, and eigenvalue analysis of the covariance matrix can reveal redundancy, collinearity, and opportunities for efficient dimensionality reduction [5,6].

For more intuitive interpretation, two-dimensional projection techniques are frequently employed. These include Principal Component Analysis (PCA)—which linearly preserves global variance—and nonlinear methods such as t-distributed Stochastic Neighbor Embedding (t-SNE), Uniform Manifold Approximation and Projection (UMAP), and Isometric Mapping (Isomap), which preserve local neighborhood structures and can uncover latent clusters in the data [7]. Moreover, separability can be empirically evaluated by training simple classifiers (e.g., k-Nearest Neighbors with $k = 1$ or 3, Linear Discriminant Analysis, or linear Support Vector Machines) without parameter tuning, as a proxy for the intrinsic discriminative capacity of the feature set [8].

- The aim of this study is to analyze and compare different types of features extracted from ECG signals, with respect to their utility in automated classification tasks. **Specifically, we investigate the performance of traditional features—such as QRS duration, QT interval, ventricular rate, atrial rate, R axis, T axis, Q onset, Q offset, and T offset—in relation to an extended set of modern features derived from the time domain, frequency (spectral) domain, and nonlinear analysis of the signal.** This comparison seeks to highlight the substantive potential of advanced features in enhancing computer-aided diagnostic systems, particularly within the frameworks of big data and machine learning. By integrating these complementary perspectives, the present study contributes to a more comprehensive understanding of how temporal, spectral, and ECG signal complexity features can be effectively combined with classical morphological characteristics—such as wave shape and onset/offset timings—to improve automated classification, facilitate early pathology detection, and support clinical decision-making.

- Key Features Used In The Paper:

In this study, two types of features were used, namely

- (i) Features related to the ECG waveform presented in ref. [11];
- (ii)

Features in the time domain, frequency, and signal-complexity measures [10].

Both types of features were extracted from the DII lead. Since the presence of noise can pose a significant obstacle to any statistical analysis, we opted for a methodology in which the first step was noise removal.

In general, the sources of noise contamination in the ECG data from this database were due to power line interference, electrode contact noise, motion artefacts, muscle contractions, baseline wander, and random noise. Therefore, to eliminate noise, we applied a band-pass filter that retained only the frequency range between 0.5 Hz and 50 Hz. This was followed by LOESS (Locally Weighted Scatterplot Smoothing) smoothing to remove baseline wander effects and subsequently by Non-Local Means (NLM), an advanced denoising method. Unlike classical filtering techniques that consider only neighboring points, Non-Local Means searches for similar patterns throughout the entire signal, not just in the vicinity. The application of this technique results in a smoothed version of the signal that preserves genuine details while eliminating noise. Thus, this method retains structural information and reduces noise without blurring signal features.

The denoising technique used is the one presented in ref. [11]. We chose to maintain the same denoising method to ensure a fair comparison of the results and because it represents a sound methodological approach.

3.5.1. Morphological Features of the ECG Waveform

The features extracted from lead II include ref. [11]:

- Ventricular rate in BPM (beats per minute),

- Atrial rate in BPM
- QRS duration in msec
- QT interval in msec
- Corrected QT interval in msec,
- R axis
- T axis
- QRS count
- Q onset (samples)
- Q offset (samples)
- T offset (samples)

3.5.2. Features in the Time Domain, Frequency and Signal-Complexity Measures

Information regarding age and sex was retained as features, given its relevance in virtually all medical data analyses.

Based on the preprocessed lead II signals, we extracted features from the time and frequency domains, along with nonlinear dynamics measures. These features were also utilized in ref. [10] for ventricular fibrillation prediction and detection, demonstrating a strong capability to capture variations in heart rhythm.

Time-domain features include both simple indicators and measures derived from the signal dynamics, such as

- amplitude range—AR,
- peak-to-peak amplitude—PPA,

- mean amplitude—MA,
- median stepping increment—MSI,
- signal integral—SignalInt,
- two variants of the root mean square (RMS) value, specific two ventricular fibrillation signals (RMS1 and RMS2)
- mean slope—MS,
- median slope—MdS,
- a smoothed nonlinear energy operator—SNEO.

Spectral features were obtained by applying the Fast Fourier Transform (FFT) using 2048 points and a Hamming window over the analysis interval. These features include

- amplitude spectrum area—AMSA,
- centroid frequency—CF,
- peak frequency—PF,
- total energy—ENRG,
- spectral flatness measure—SFM,
- centroid power—CP,
- maximum power—MP,
- power spectrum analysis—PSA.

To assess signal complexity, we also included measures based on nonlinear dynamics. Among these are

- Hurst exponent (Hu) ,
- scaling exponent—ScE,
- logarithm of absolute correlations—LAC ,
- the median increment derived from the Poincaré plot (MSI).

In addition, two entropy-based metrics were calculated:

- entropy—WE,
- spectral entropy—SpeEnt.

Paper 10: Heart arrhythmia diagnosis based on the combination of morphological, frequency and nonlinear features of ECG signals and metaheuristic feature selection algorithm

Link: [Heart arrhythmia diagnosis based on the combination of morphological, frequency and nonlinear features of ECG signals and metaheuristic feature selection algorithm - ScienceDirect](#)

My Findings From The Paper - Key Points Worth Noting:

- This paper extracts features including R-wave amplitude, QRS complex depth, previous and subsequent RR intervals, power spectral density via Welch's method and discrete Fourier transform, and HRV signal features including sample entropy, Poincaré plot, and spectral features of high and low-frequency bands.
- In this study, a new method for arrhythmia discriminating and classifying the ECG signals is proposed based on a combination of morphological, frequency, and nonlinear feature extraction methods. After the segmenting the ECG signal, the duration, amplitude, variance and mean of P and R waves, average and standard deviation of RR intervals, QRS complex interval, average energy and power of samples as the morphological characteristic, PSD and DWT as the frequency features and fractal dimension (FD), Lyapunov exponent (LE), approximate, sample, fuzzy and permutation entropy and Poincare plot as the nonlinear indices are computed for the dataset signals.

Paper 11: Multilabel 12-Lead Electrocardiogram Classification Using Gradient Boosting Tree Ensemble

Link: [\[2010.13712\] Multilabel 12-Lead Electrocardiogram Classification Using Gradient Boosting Tree Ensemble](#)

My Findings From The Paper - Key Points Worth Noting:

- The extracted features include heart rate variability features, PQRST shape templates, autoregressive model coefficients, change quantiles, aggregate linear least-squares regression trends, peak counts, sample/approximation entropy, energy, continuous waveform transform coefficients, fast Fourier transform coefficients, and other descriptive statistics.

Other Papers Worth Looking At:

- [A New 12-Lead ECG Signals Fusion Method Using Evolutionary CNN Trees for Arrhythmia Detection](#)
- From UCI Machine Learning Dataset For Arrhythmia Prediction: [pxc3878532-libre.pdf](#)
- Under “Methods” section: [Prediction of mortality from 12-lead electrocardiogram voltage data using a deep neural network | Nature Medicine](#)
- Based on residual structure and attention mechanism: [An Automated Cardiac Arrhythmia Classification Network for 45 Arrhythmia Classes Using 12-Lead Electrocardiogram | IEEE Journals & Magazine | IEEE Xplore](#)